

EVA-1 from KK System – Reproduction

User Manual

v1.1

1 Building

1.1 SMD or THT Components

There are several components with dual footprint (SMD or THT). The Reference for SMD is THT+100. So Capacitor C1 is a THT part but if one wants a SMD part it is C101. The same goes for some Resistors (R13, R113, R17, R117,...). It should be clear to assemble only one component!

1.2 Floppy Extension

The floppy extension uses an Arduino Mega 2560. Unfortunately the Arduino Pro micro doesn't work, it crashes while sending the boot extension. The firmware is PFBDK with the HX20 Extension. ([PFBDK Github Site](#)). There are some parts which don't need assembly when not using this feature.

1.3 VGA Output

There is space for a Raspberry Pi Pico (RP2040) on the board. It converts the CGA Type Video into a VGA Compatible Signal. The firmware on the Pi Pico is MCE Blaster ([MCE Blaster Github Site](#)).

There is no connection by default to the Pi Pico, one must connect J13 to J12 according to the wanted colour since the Output of the EVA1 is only monochrome.

1.4 CPU

There is the possibility to use two different CPUs. Default is a MC6803R, which is the 1MHz Variant in a DIP-40 Package. The MC68A03 and MC68B03 are the 1.5MHz and 2MHz Types. Although the CPU runs at slightly over 1.5MHz (1.54 to be precise) all tested CPUs worked. The MC68B03R is the preferred Version. If one has a Hitachi CPU (HD63x03R) there is a Solder Jumper on the back (JP6) which needs to be open since the HD6303 needs Pin 2 open while the MC6803 needs it on GND (see datasheet).

R17 and R13 are important and must be placed! They select the CPU Mode.

1.5 Firmware ROM

The firmware is only 4kB, so a 2732 would be enough but since the 2732 has only 24pins and is difficult to program compared to a 2764 or 2864 the pcb has an 28pin socket which can support any EPROM from 64kbit to 512 kbit. For the prototype an AT28C64 EEPROM has been used. For this type JP10 has to be open (default). For an EPROM JP10 should be closed. With a 27C256 one can connect a switch to Pin 27 (A14) and GND and use two different firmware variants.

1.6 Font ROM

The Font ROM uses a 2732 EPROM. There is an alternate Socket for a 2764 with a slightly different pinout. There is however no modified Font ROM file for a 2764 at the moment.

1.7 GALs

There are two GALs on the PCB (U32 and U45). The .jed files are in the GAL folder. The TL866 can program the gals just fine.

1.8 HD6845

Only the Hitachi HD6845 is supported. The MC6845 has the ouptut shifted to the left.

1.9 Clock

There are two footprints for the 18.432 MHz Clock. One DIP-8 and one DIP-14. Use whatever. No Crystals are supported only MXO Type Oscillator.

1.10 74 Series Variants

The prototype is tested with LS and HCT variants and for the most parts both are fine except for the '157 multiplexer. Only the 74 ALS 157 have been tested.

1.11 Various Jumpers

JP8 can be closed if only a “straight” cable is at hand. In this case use the right DIN-5 connector. An external Floppy (TF-10 for example) can NOT be used! The internal Arduino emulator would work fine.

JP9 is for double speed operation. One can “overclock” the HX-20 with a faster crystal (two 4.9152 MHz needed). In this case all communication takes place at double speed (38400 Baud * 2). The Arduino firmware has to be changed as well!

1.12 Case

The case is a Strapubox 5003. Supplier is Segor electronics.

2 Connection

2.1 Serial connection:

The connection between HX-20 and EVA-1 must be with a 5-pin DIN “crossover” cable. Pinout of the cable:

Pin HX20	Pin EVA1	Function	Comment
1	1	GND	
2	3	PTX Out (Transmit Data)	

3	2	PRX In (Receive Data)	
4	5	Pout Out (Transmit Mode)	not used
5	4	Pin In (Receive Mode)	not used

The cable must be connected to the left socket. The right socket is for a portable floppydrive.

In the new Version of the Board is a jumper (JP8). When this Jumper is closed, a “straight 5-pin DIN cable can be used. This cable must be connected to the right socket. In this case a portable floppydrive cannot be used!

2.2 Monitor Connection:

There are two monitor connections, a Video connector (FBAS, SCART compatible) and a CGA Type connection (15.6kHz, 50 Hz) for a CGA compatible monitor.

3 Usage

After connection the monitor and power the startup screen should show. Here are the most common shortcuts and also the first entry for the HX-20.

3.1 Control characters:

CTRL+E	Deletes from Cursor to the end of the line
CTRL+F	Toggle Inverse (only for “special” special characters)
CTRL+G	Toggle bright (only for “special” special characters)
CTRL+N	“special” character from now on. This writes every character with Bit 7 set. It won’t show unless one “activates” them with CTRL+F or CTRL+G
CTRL+I	Horizontal TAB
CTRL+P	Move Window up. The TEXTRAM Chip on the Reproduction board is twice as big. It can store 50 Lines with 80 characters. Moves only the Window of the screen buffer.
CTRL+Q	Move Window down. Moves only the Window of the screen buffer.
CTRL+T	Restores the original screen buffer window.
CTRL+S	Jumps into monitor program. In the new EPROM is a monitor program (MC3 monitor from Daniel Tufvesson). After CTRL+S connect a terminal (38400Baud) and press Enter.
CTRL+V	Cursor visible
CTRL+W	Cursor invisible
CTRL+Z	Deletes from Cursor to the end of the screen
CTRL+0	Show test picture (works only in SCREEN 1,1 Mode)
CTRL+1	toggle Text ON/OFF
CTRL+2	toggle Inverted screen
CTRL+3	switch to 32 characters / Line
CTRL+4	switch to 40 characters / Line
CTRL+5	toggle Cursor BLINK
CTRL+6	Clear graphics screen (GCLS)
CTRL+7	switch to 72 characters / Line
CTRL+8	switch to 80 characters / Line
CTRL+9	toggle character set (the second character set has only minor differences – see Reference Manual Appendix F)

4 EPSP Functions

The EVA-1 handles the following EPSP Functions:

\$88	SCRNSIZE	READ SCREEN SIZE (L.C)
\$89	SCRNSIZE	GET PHYSICAL SCREEN SIZE (L.C)
\$8C	GCRSRPOS	GET CURSOR POSITION
\$8F	GETPOINT	GET POINT ON THE DISPLAY (L.C)
\$91	READLINE	READ THE EXTEND OF CURRENT LINE (L.C)
\$92	WRITECHAR	WRITE ONE CHARACTER TO VIRTUAL SCREEN (L.C)
\$93	SETDISPMOD	SET DISPLAY MODE (C)
\$97	READCHAR	READ CHARACTERS FROM VS. (L,C)
\$98	WRITECHAR	WRITE ONE CHARACTER TO VS AND GET EXTEND OF NEW CURSOR POSITION.(L,C)
\$C2	SETCRSR	SET CURSOR POSITION ON THE VIRTUAL SCREEN (L.C)
\$C5	SRLISTFLG	SET LIST FLAG (L.C)
\$C6	SRLISTFLG	RESET LIST FLAG (L.C)
\$C7	SETPOINT	SET POINT TO DISPLAY (L.C)
\$C8	DRAWLINE	DRAW LINE TO DISPLAY (L.C)
\$C9	SETLINETERM	SET LINE TERMINATE POSITION (L.C)
\$CA	CLRGRAPH	CLEAR THE GRAPHICS SCREEN. (L.C)
\$CB	DISABLED	SET SCRLUPSPEED (C)
\$CF	DISABLED	COLOR SET SELECT (C)
\$D4	DISABLED	?